

Math
for
EE

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$$\begin{aligned} I_{rms} &= \\ \frac{V_{rms}}{Z} &= \\ \frac{V_{rms}}{R} &= 21.4 \\ \frac{V_{rms}}{R} &= 240V, R = \\ 60\Omega &= 21.5 \\ \frac{V_{rms}}{Z} &= \\ 4.4 &= 21.6 \\ x^3 - & \\ 2x - & \\ 5 &= \\ 0 &= \\ x_0 &= \\ 3 & \\ x_1 & \end{aligned}$$

Solution:

$$\begin{aligned} f(x) &= \\ x^3 - & \\ 2x - & \\ 5 &= \\ x &= \\ 222.1 &= \\ \frac{x_n}{n+1} &= \\ \frac{f(x_n)}{f'(x_n)} &= 22.2 \\ 0 &= \\ x_0 &= \\ 3 & \\ f(x_0) &= \\ 3 &= \\ 2(3) - & \\ 5 &= \\ 37 &= \\ 6 &= \\ 5 &= \\ 16 &= 22.3 \\ \frac{f'(x_0)}{2} &= \\ 2 &= \\ 3(9) - & \\ 2 &= \\ 27 &= \\ 2 &= \\ 25 &= 22.4 \\ x^1 &= \\ 1 &= \\ 3 &= \\ \frac{16}{2^5} &= \\ 3 &= \\ 0.64 &= \\ 2.36 &= 22.5 \\ \sigma &= \\ \sigma_0 &= \\ 2 \log_e(t) & \text{fort} \geq \\ 1 &= \\ \sigma_0 &= \\ t &= \\ 2MPa &= \\ 4 &= \\ MPa &= \\ 8 & \end{aligned}$$

Solution:

$$\begin{aligned} \sigma(t) &= \\ t &= \\ \sigma(t) &= \\ \sigma_0 &= \\ 2 \ln(t) &= 23.1 \\ t &= \\ 4 &= \\ \sigma(4) &= \\ 2MPa &= \\ \sigma_0 &= \\ 2 \ln(4) &= 23.2 \\ t &= \\ 8 &= \\ \sigma(8) &= \\ \sigma(8) &= \\ \sigma_0 &= \\ 2 \ln(8) &= 23.3 \\ \sigma(8) &= \\ 2 &= \\ -2 \ln(8) &= \\ (-2 \ln(4)) &= 23.4 \\ \sigma(8) &= \end{aligned}$$